LAB REPORT

CSE312: Database Management System Lab

Submitted To
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(Lecturer)
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Submitted By
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Section**: 66_L2**
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Date of Lab Report Submission: 05 February 2026

Experiment No: 01	Experiment Name: **Design and Implementation of Hospital Database Using SQL DDL.**

1. Experiment Details:

Create a database named **Hospital** and design the following tables:

- **Doctor** {d_id(pk), d_name, specialist, **room_no(fk)**}
- **Patient** {p_id (pk), p_name, diagnosed_with, **room_no (fk), assigned_doctor(fk**)
- **Room** {**room_no(fk),** type (ward, cabin, vip cabin, doctor_chamber)}

All key and required fields must be defined using the **NOT NULL** constraint.

2. Software & Tools Used:

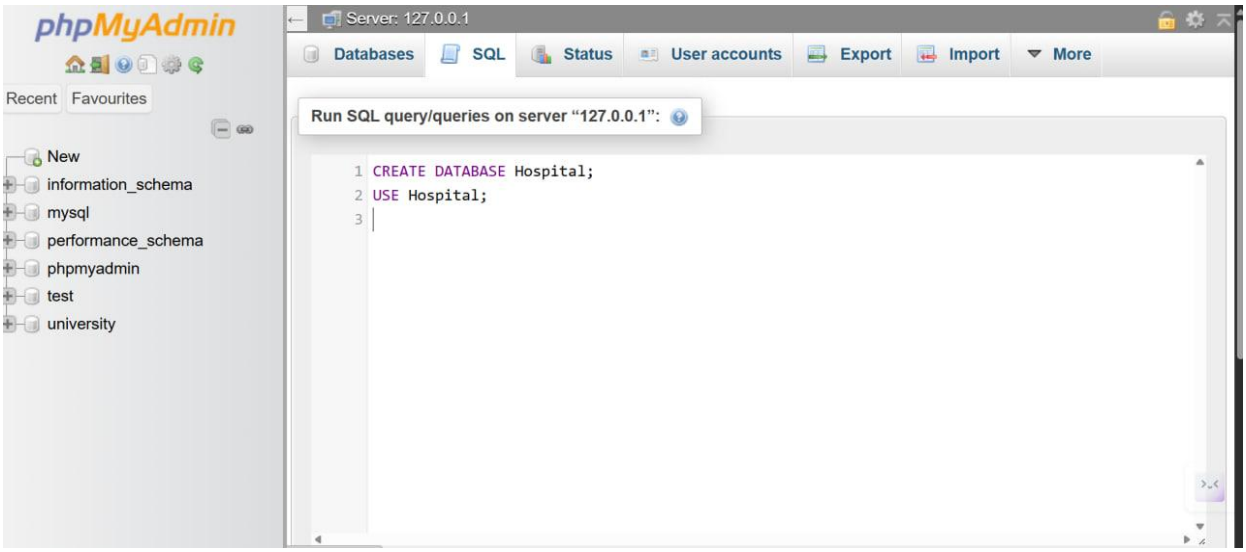
- Web Server Stack: XAMPP Control Panel v3.3.0
- Database Management System: MariaDB 10.4.x (MySQL-compatible)
- Database Server: MySQL Server (running on localhost via XAMPP)
- Interface Tool: phpMyAdmin 5.2.x
- Web Server: Apache 2.4.x
- Scripting Language: PHP 8.0.x
- Operating System: Windows 11

3. Methodology:

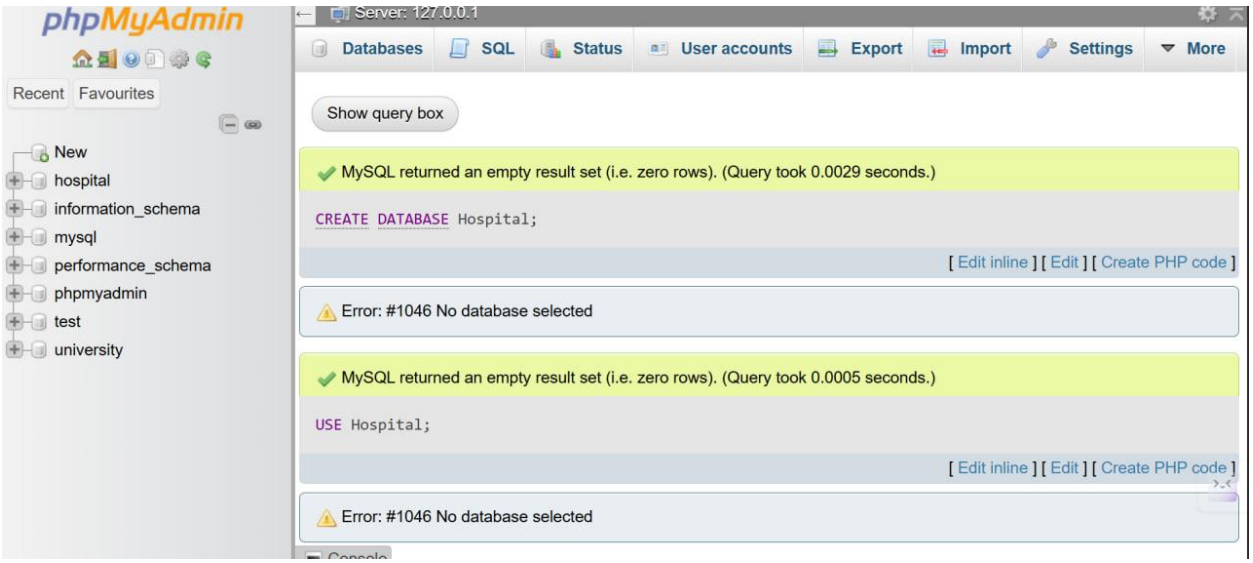
- Created the **Hospital** database using the CREATE DATABASE command.
- Selected the database using the USE command.
- Created the **Room** table as the parent entity.
- Created the **Doctor** table with a foreign key reference to the Room table.
- Created the **Patient** table with foreign key references to both Room and Doctor tables.
- Applied NOT NULL constraints to all key and mandatory attributes.
- Verified table creation and structure using SQL queries.

4. Experiment Solutions:

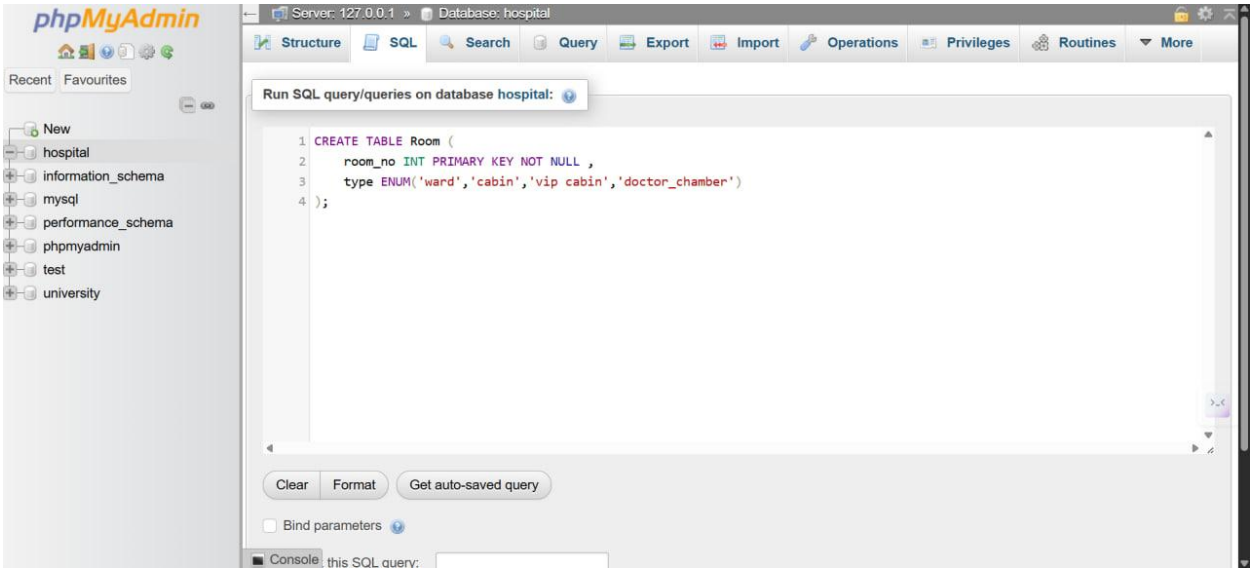
Step-01: CREATE DATABASE (Hospital)



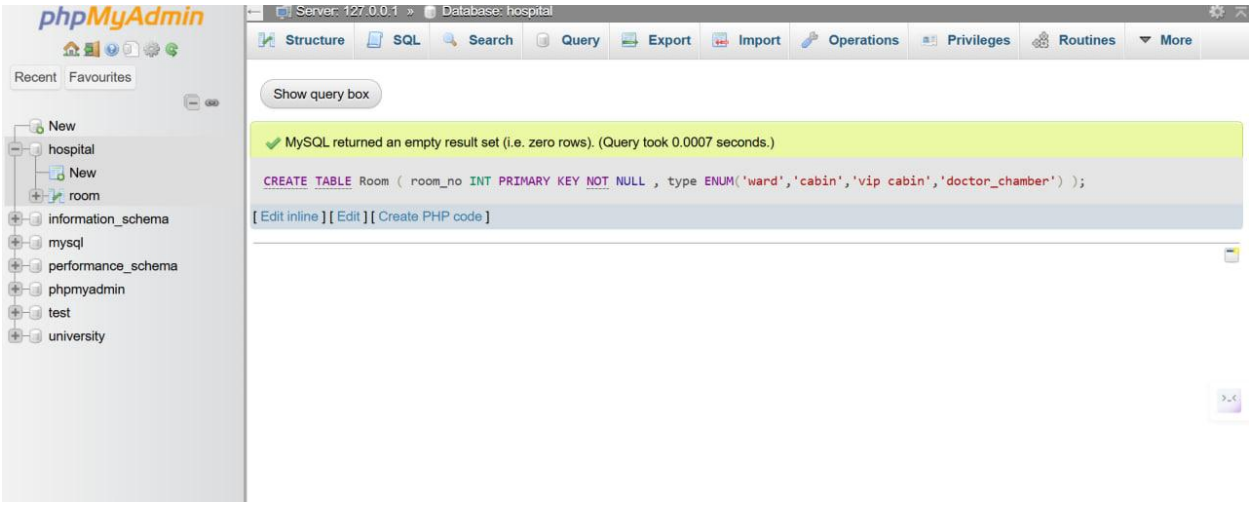
Step-02: Success of Compilation (Hospital)



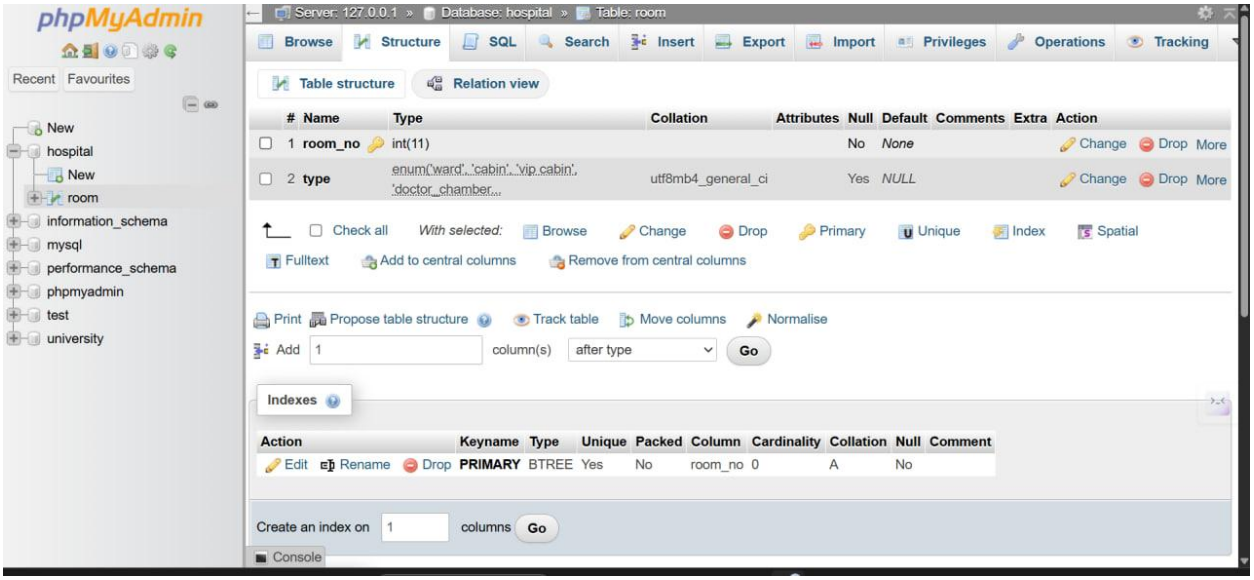
Step-03: SQL Queries (Create Room Table)



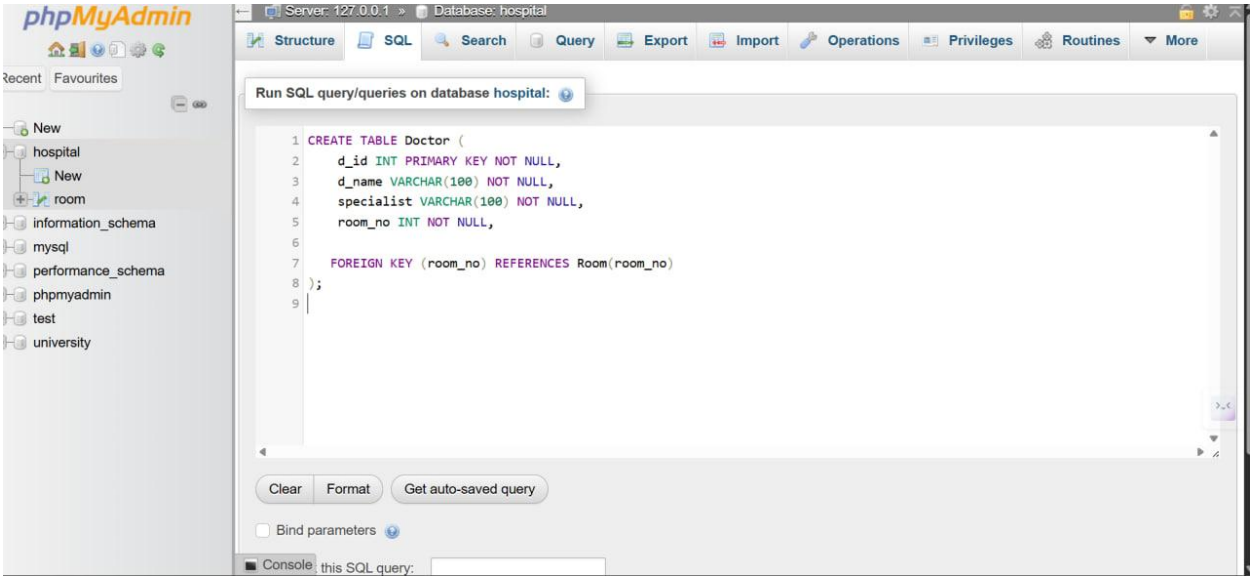
Step-04: Success of Compilation (Room Table)



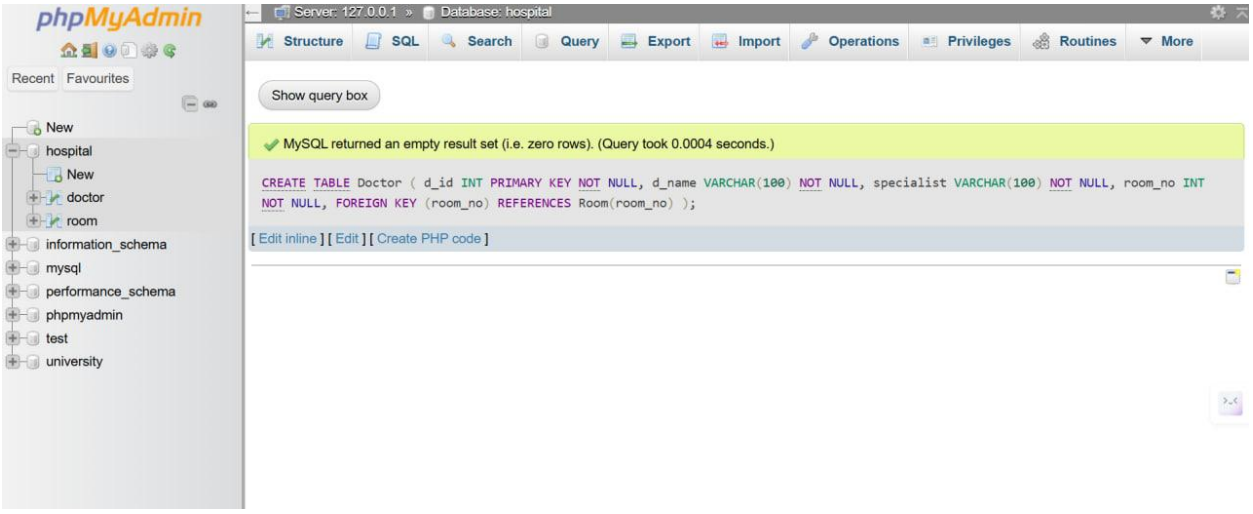
Step-05: Output (Room)



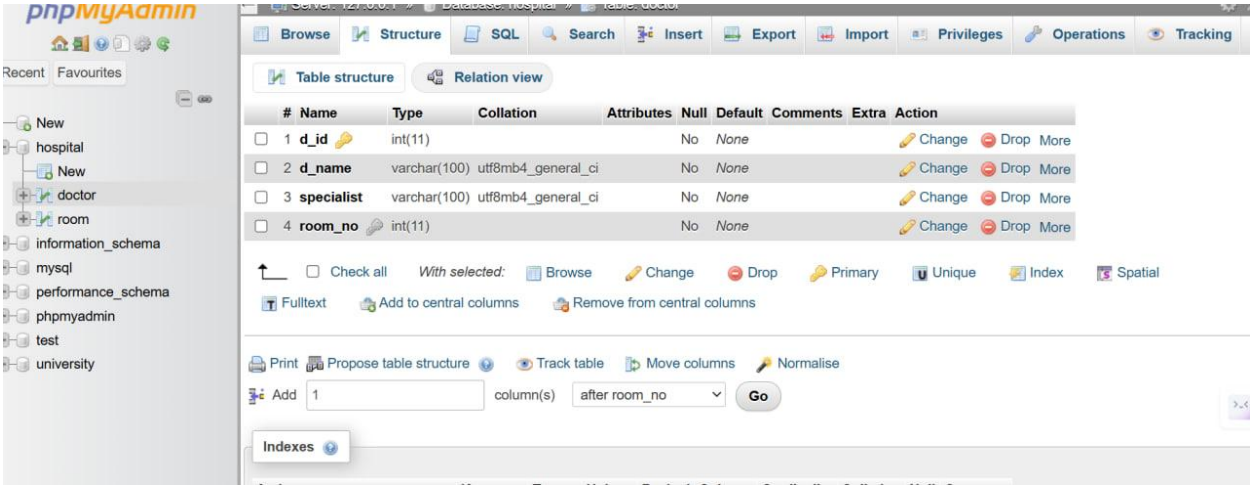
Step-06: SQL Queries (Create Doctor Table)



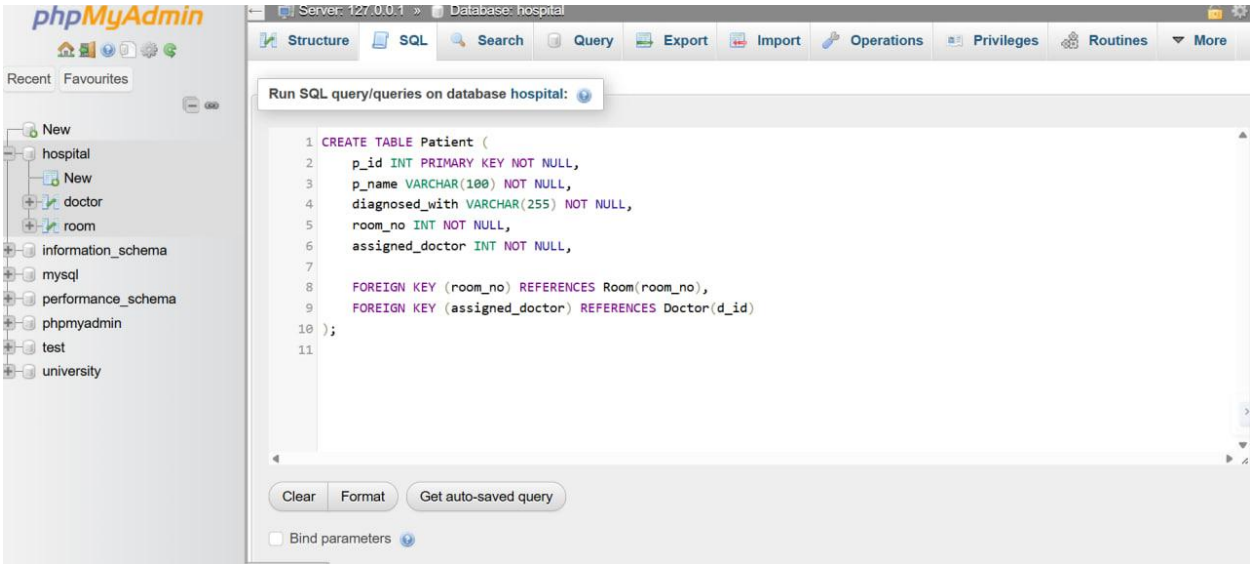
Step-07: Success of Compilation (Doctor Table)



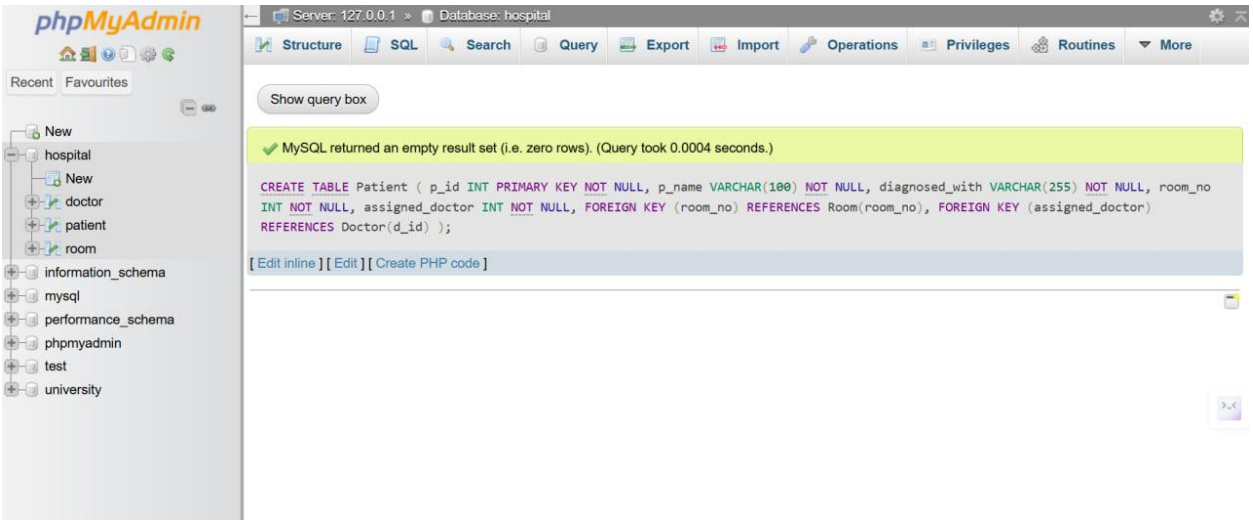
Step-08: Output (Doctor)



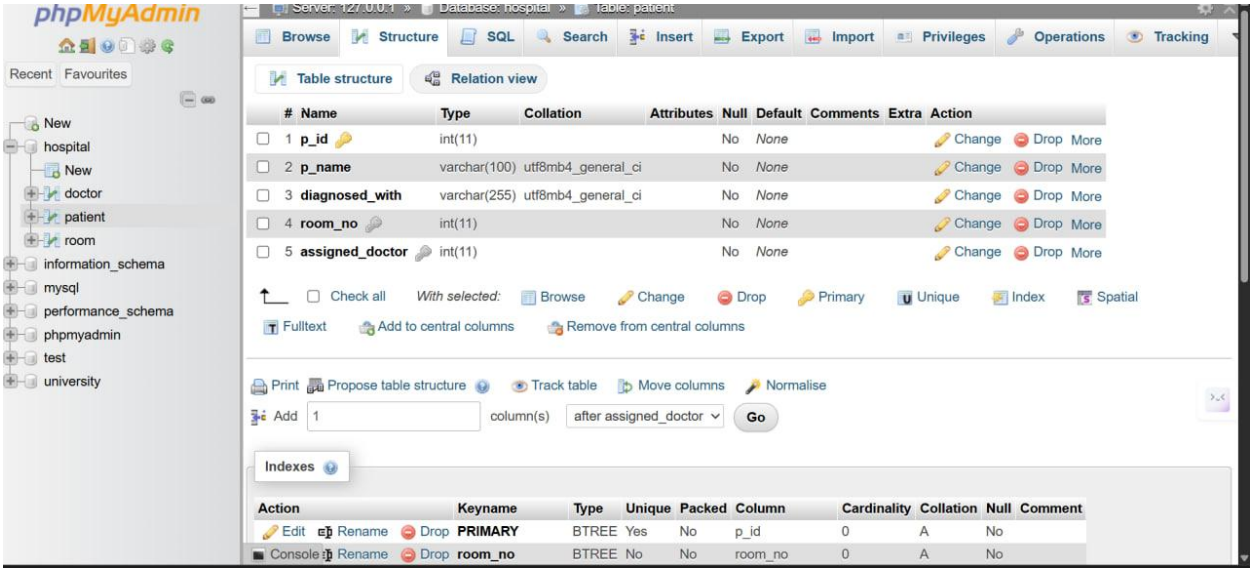
Step-09: SQL Queries (Create Patient Table)



Step-10: Success of Compilation (Patient Table)

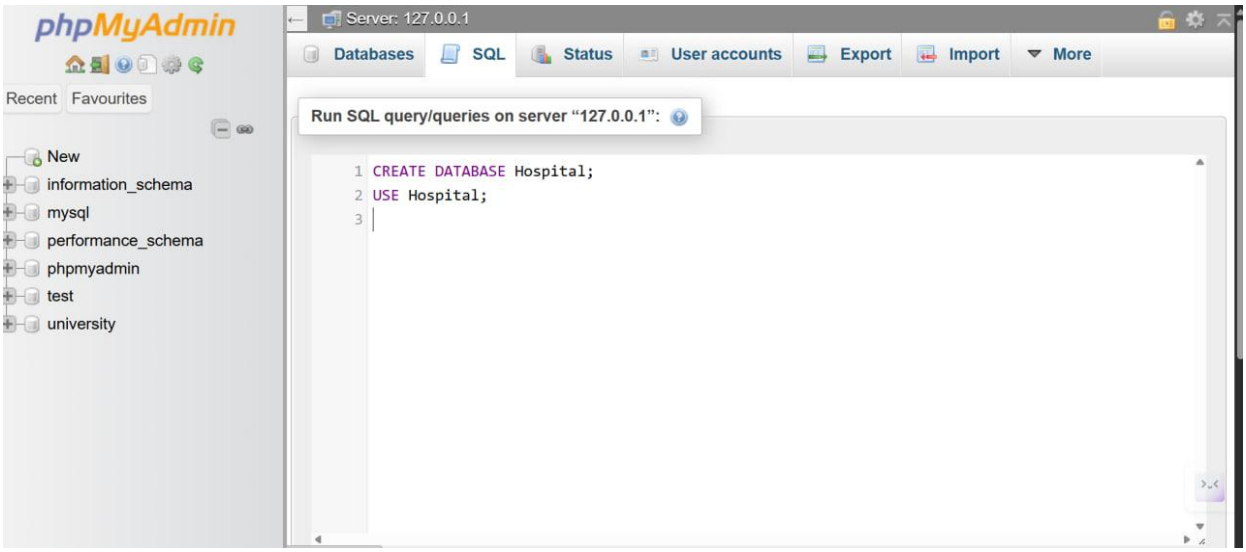


Step-11: Output (Patient)

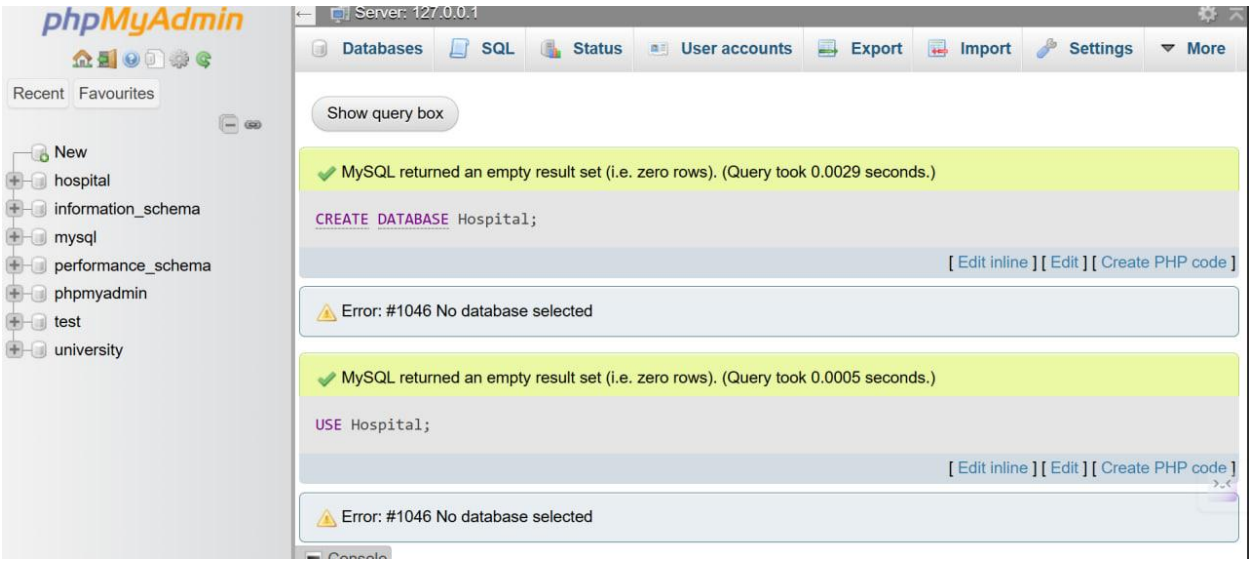


4.1. Alternative Steps:

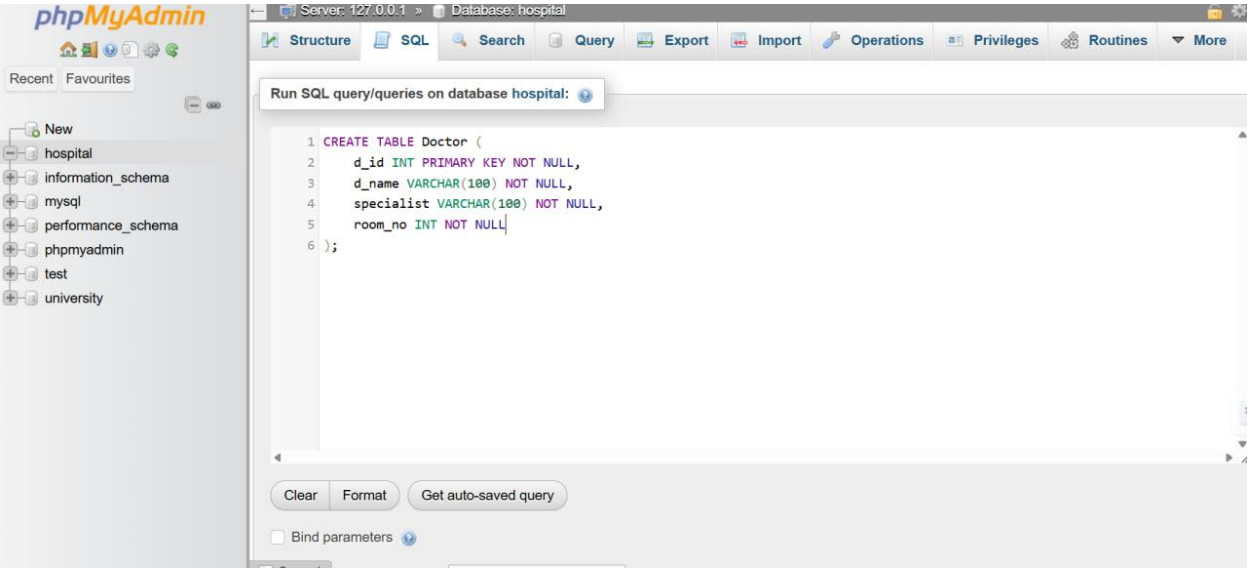
Step-01: CREATE DATABASE (Hospital)



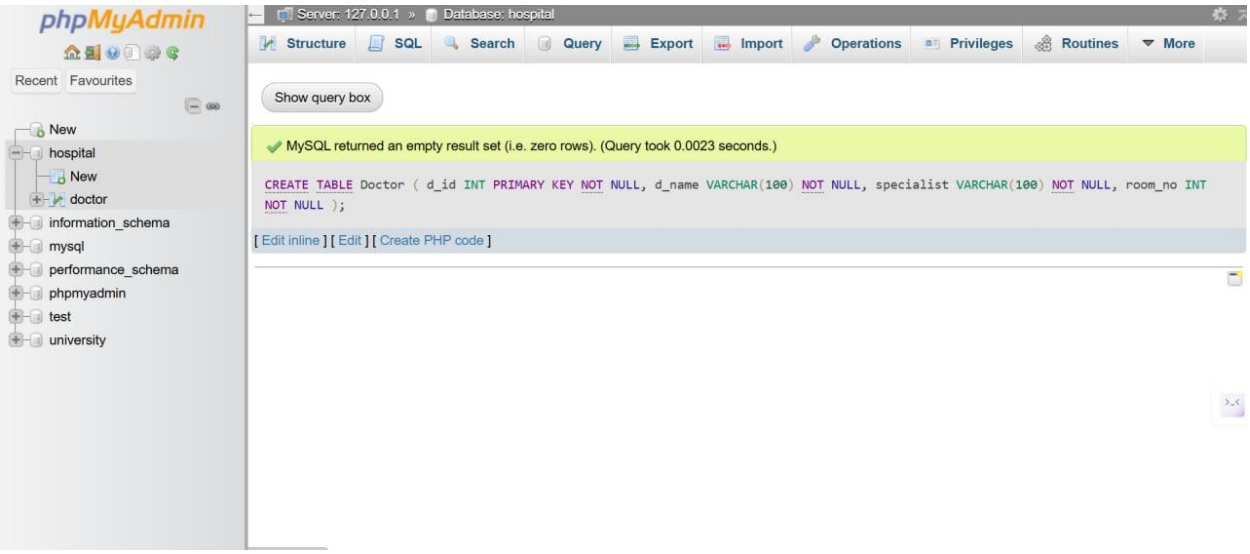
Step-02: Success of Compilation (Hospital)



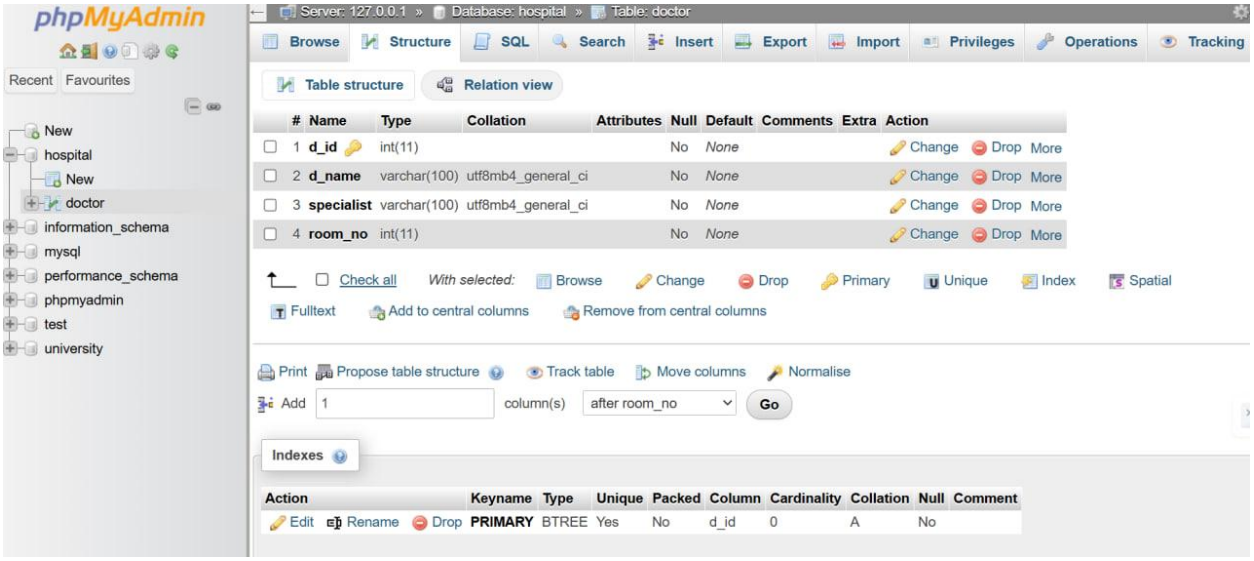
Step-03: SQL Queries (Create Doctor Table)



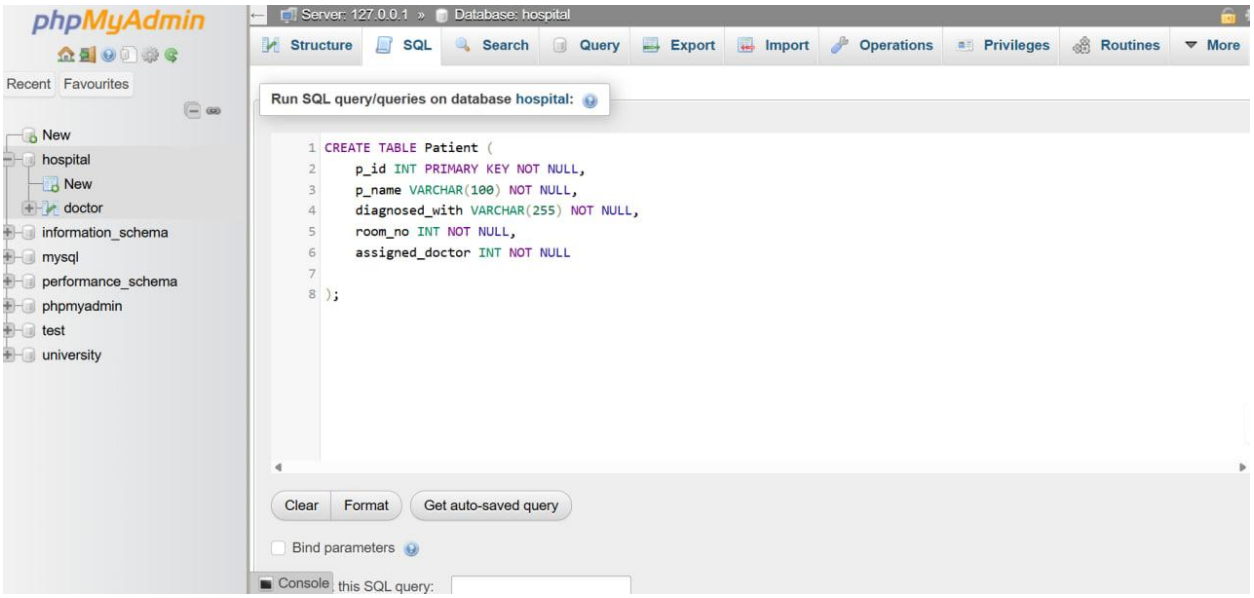
Step-04: Success of Compilation (Doctor Table)



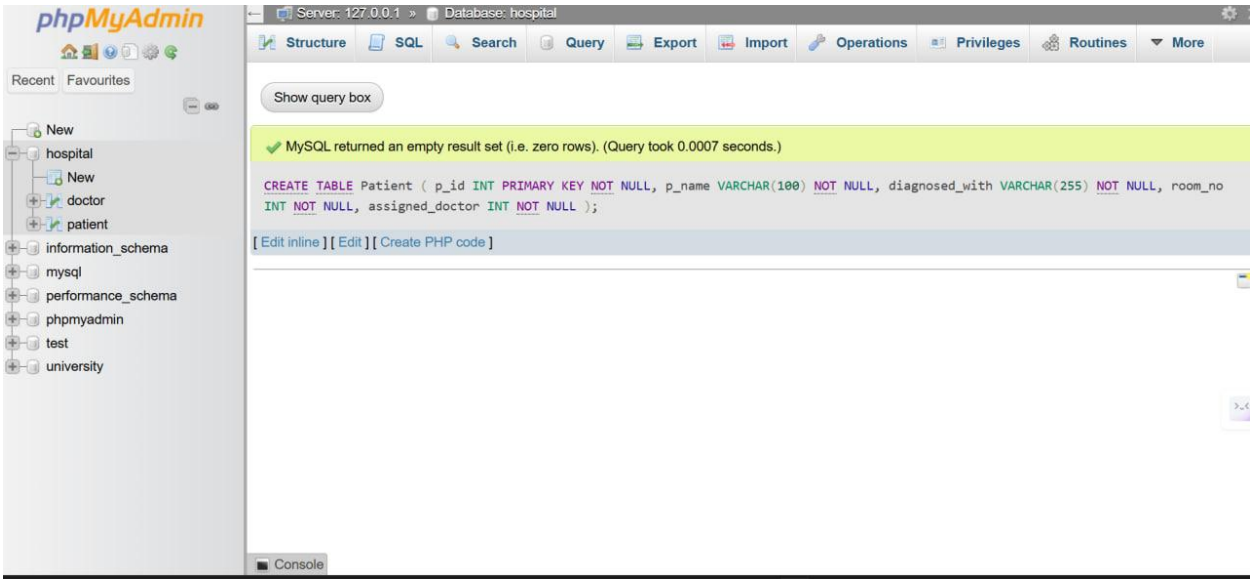
Step-05: Output (Doctor Table)



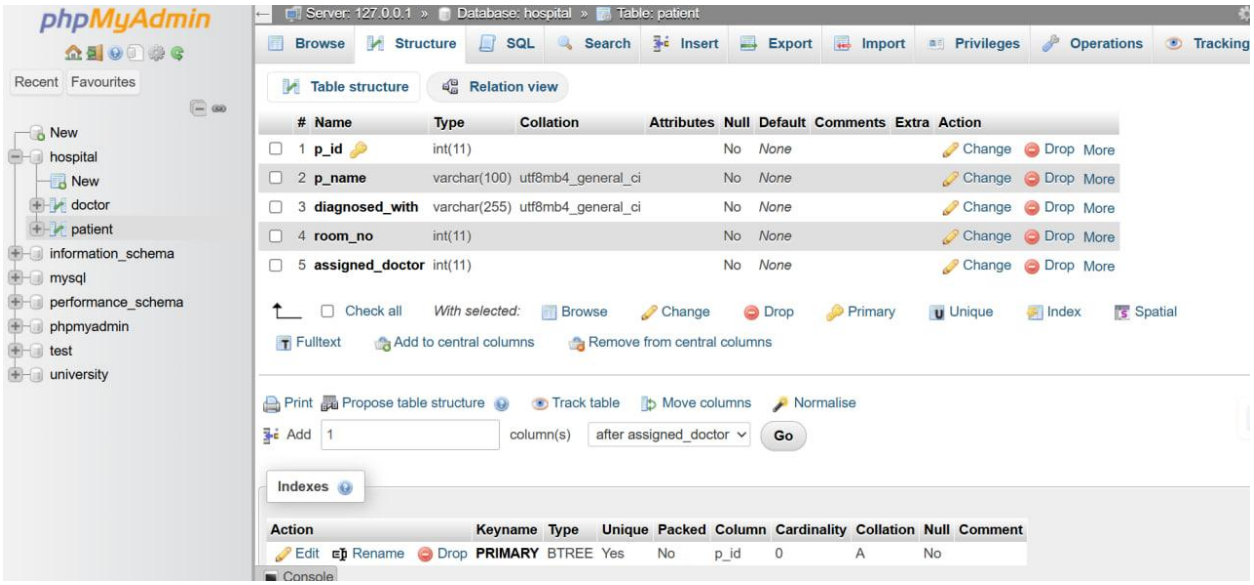
Step-06: SQL Queries (Create Patient Table)



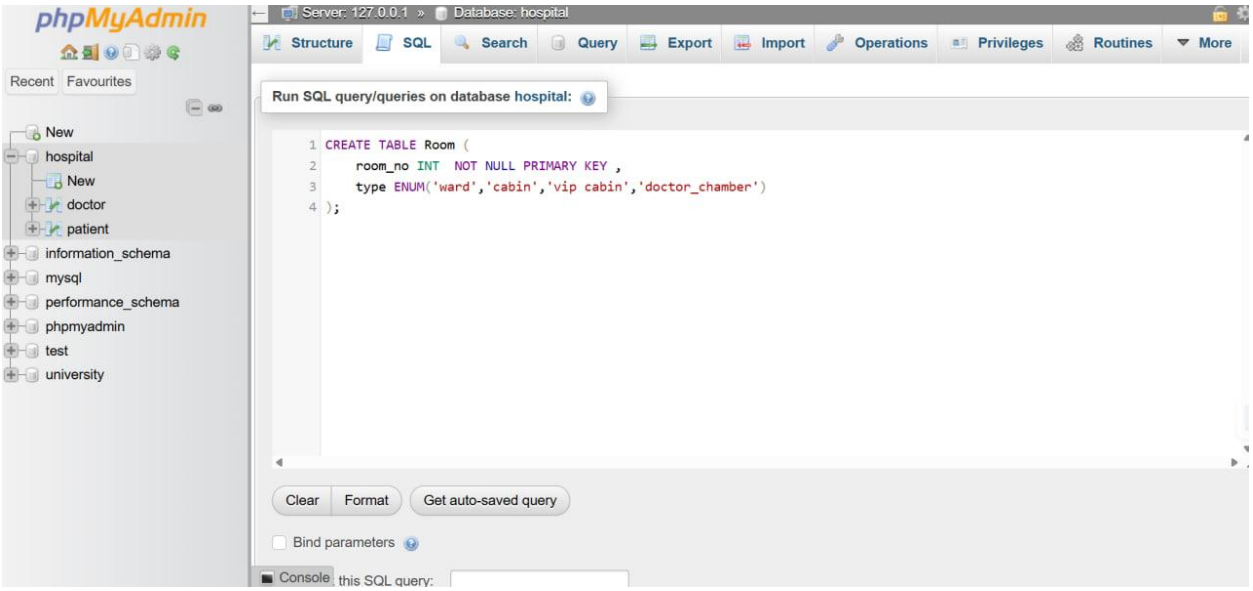
Step-07: Success of Compilation (Patient Table)



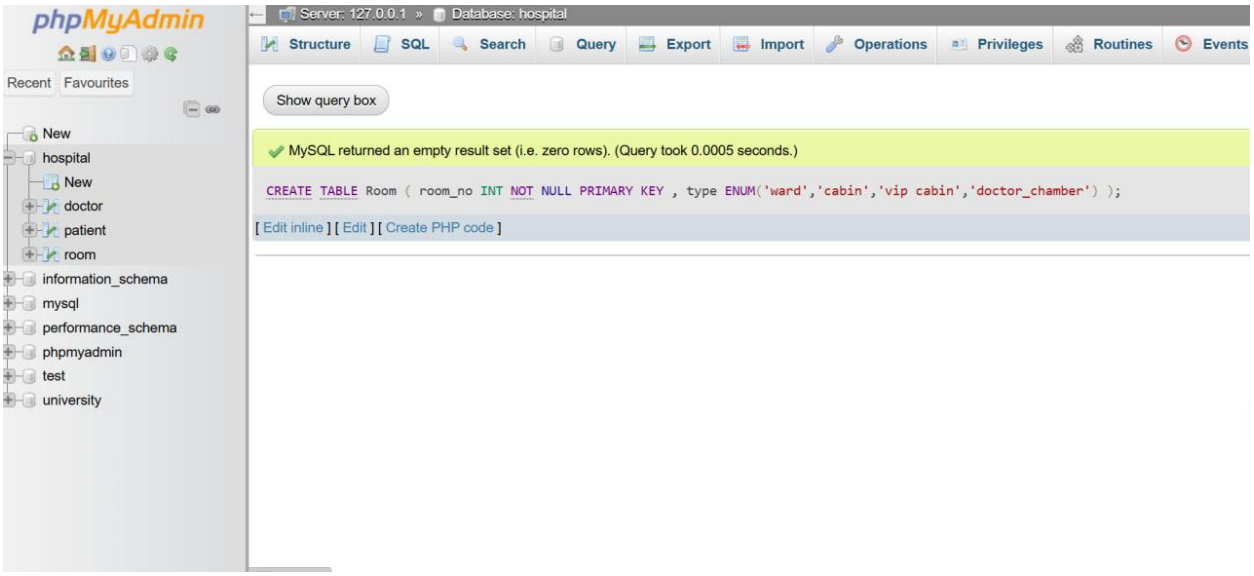
Step-08: Output (Patient Table)



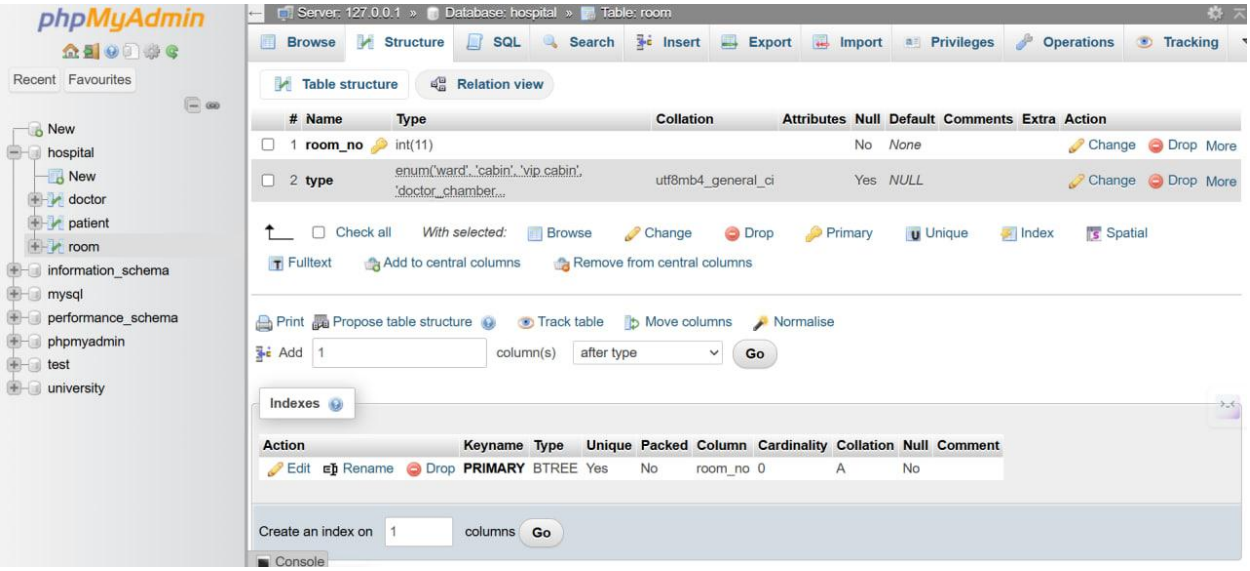
Step-09: SQL Queries (Create Room Table)



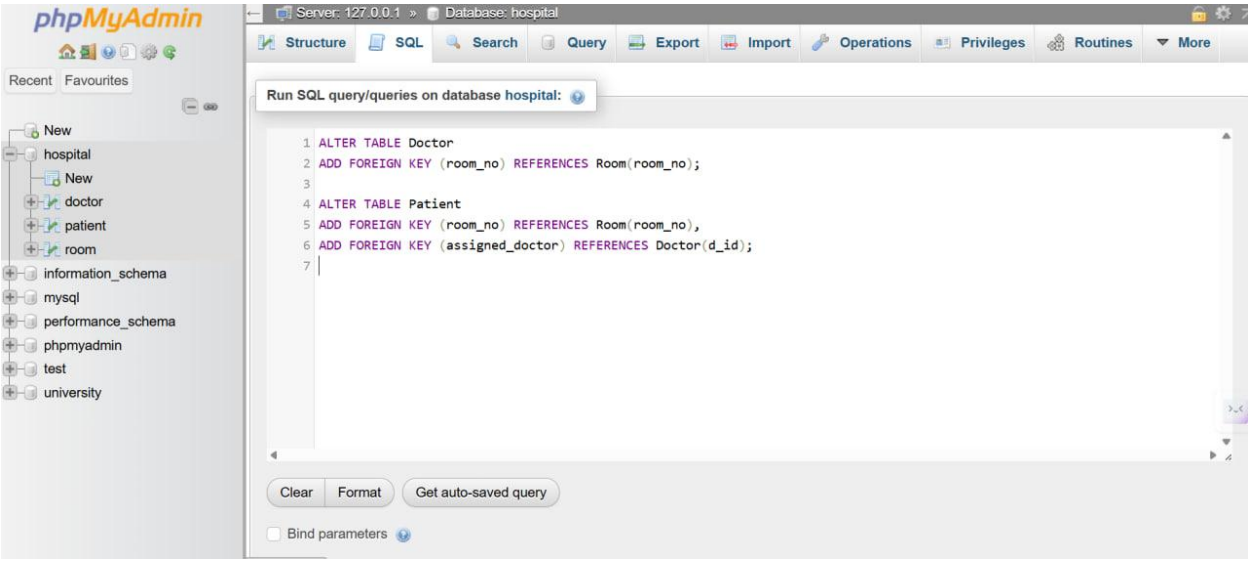
Step-10: Success of Compilation (Room Table)



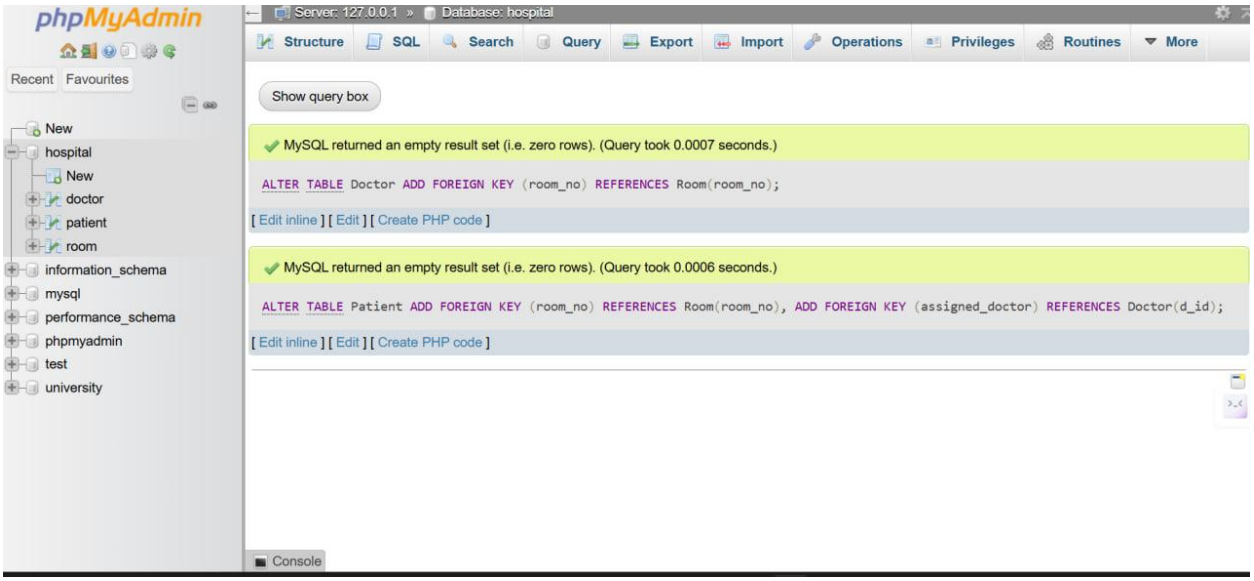
Step-11: Output (Room Table)



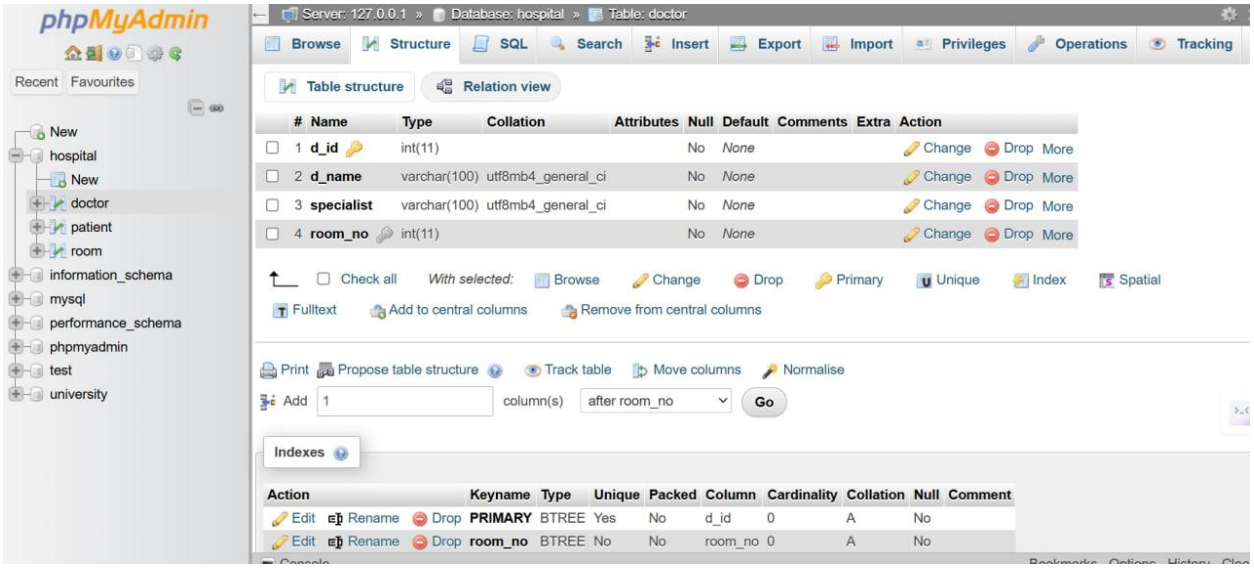
Step-12: SQL Queries (Add foreign key constraints after table creation)

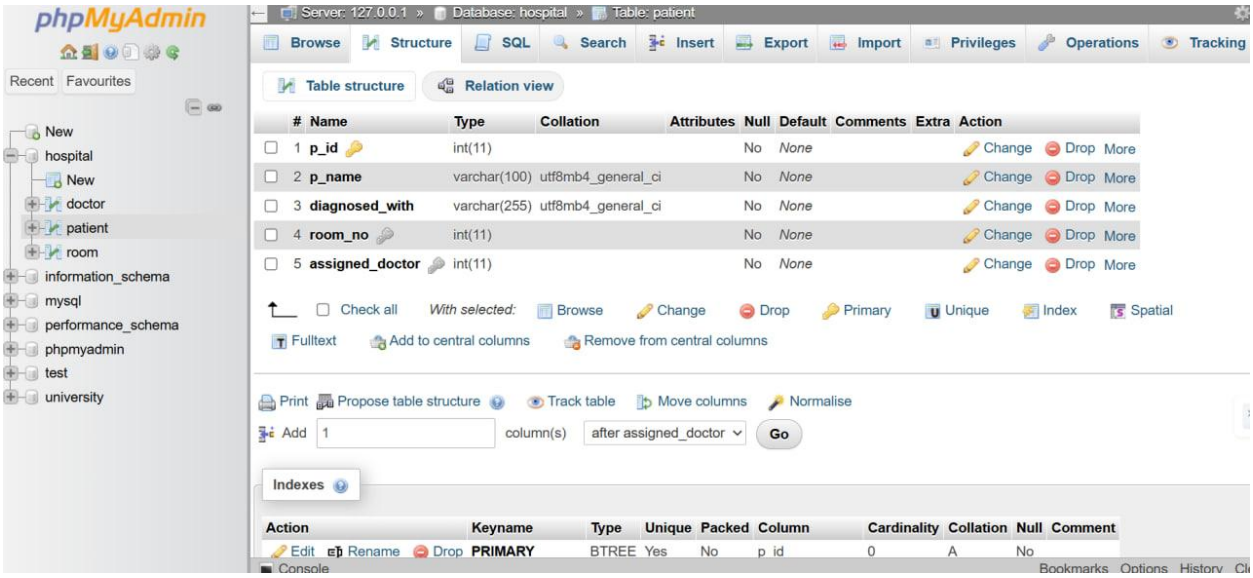


Step-13: Success of Compilation



Step-14: Output





5. Observation:

- Primary and foreign key constraints were correctly applied.
- NOT NULL constraints ensured that all mandatory fields contain valid data.
- Referential integrity among the tables was successfully maintained.
- The experiment improved understanding of relational database schema design.

6. Appendix:

Appendix A: Course Outcomes, Complex Engineering Problems (EP) and Complex Engineering Activities (EA) Addressing.

Appendix A: Course Outcome (CO) Attainment

SN	COs	Attainment	Justification
01	CO1	Yes	These lab activities attain CO1 by practicing database creation and a couple of tables having at least 4-6 parameters and an interconnectedness between tables using primary key and foreign keys.
02	CO2	Yes	Implements relational integrity using keys and constraints.
03	CO3	No	Security and distributed DB topics are outside the scope.

Appendix B: Addressing Complex Engineering Problems (EP)

SN	Engineering Problem (EP)	Attainment	Related (CO)	Justification
01	EP1: Depth of Knowledge	Yes	CO1, CO2	Proper schema design and constraint usage.
02	EP2: Conflicting Requirements	No	CO2	Single-scope academic task.
03	EP4: Familiarity of Issues	Yes	CO1	Standard DBMS lab problem.
04	EP7: Interdependence	Yes	CO2	Inter-table dependency via foreign keys.